Skin Lesion Classification with ABC Features and Shortcut Detection

Projects in Data Science - Spring 2026

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GitHub repository: https://github.com/mihnea2020/2026-PDS-Fox.git

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Abstract

This project investigates how classical image processing and simple machine learning methods can be used to classify skin lesion images as cancerous or non-cancerous. We use the PAD-UFES-20 dataset and group basal cell carcinoma, melanoma, and squamous cell carcinoma as cancerous, while actinic keratosis, melanocytic nevus, and seborrheic keratosis are treated as non-cancerous. The baseline method extracts interpretable ABC-style features from lesion masks, including asymmetry, border irregularity, lesion area, and color variation. We then investigate shortcuts, especially hair and pen marks, by comparing automatic shortcut-density features with manual annotations and by testing an extended classifier that adds these shortcut features. The final feature table contains 117 annotated images, balanced between 59 cancerous and 58 non-cancerous cases. Across patient-aware 5-fold cross-validation, the cleaned (hair-removed) logistic regression model performs best with 73.50% accuracy and an AUC of 0.7870. The baseline logistic regression model obtains 71.79% accuracy and an AUC of 0.7744, while the extended model obtains 69.23% accuracy and an AUC of 0.7516. The drop in performance in the extended model suggests that the automatic shortcut features were noisy rather than helpful in the final subset. This is supported by low Cohen's Kappa values between automatic detections and manual labels: 0.0021 for hair and 0.3067 for pen marks. Overall, the project shows that interpretable features can be useful, but mask quality, small sample size, and shortcut-detection reliability strongly affect the final model.

Area	**Main responsible member(s)**
Introduction, related work, asymmetry, future improvements	Hafsa
Data, feature extraction support, annotation analysis	Evelina
Mask post-processing, baseline classification, extended classification, hair removal	Mojtaba Ahmadian
Feature extraction, color analysis, evaluation/results, conclusion	Mihnea-Ioan Iancu

1 Introduction

Skin cancer is one of the most common cancer-related health concerns, and early detection is important because suspicious lesions are easier to treat when they are identified early. Dermatologists often inspect lesions using visual criteria such as asymmetry, border irregularity, color variation, and diameter or differential structures. These ideas are often summarized by the ABCD rule of dermatoscopy [1]. Although this rule is designed for human experts, it also gives a useful starting point for a data science project: we can convert visual properties of the lesion into numerical features and train a classifier.

The aim of this project is to build and evaluate a complete but understandable pipeline for skin lesion classification. Instead of using a deep learning model, we use hand-crafted image features and simple classifiers. This makes it possible to inspect each part of the pipeline: image and mask loading, mask cleaning, feature extraction, shortcut detection, classification, and evaluation. We focus on the binary task of classifying lesions as cancerous or non-cancerous.

A second goal is to study shortcuts. In medical imaging, a shortcut is a pattern that may help a model predict labels on one dataset but is not actually the medical signal of interest. For skin lesions, examples include hair, pen markings, rulers, lighting, and camera artifacts. Hair is especially relevant because it can cover the lesion, create strong edges, and influence measured color features. Pen marks can also introduce artificial color and border information. Our project therefore asks:

> *Do automatic shortcut features for hair and pen marks improve a simple ABC-style skin lesion classifier, or do they add noise?*

The report is organized as follows. We first describe the dataset, annotations, and label grouping. We then explain mask post-processing, baseline features, shortcut detection, hair removal, and the classifiers. Finally, we report the results, analyze why the extended model did not improve performance, and discuss limitations and future improvements.

2 Related Work

Classical skin lesion analysis often uses hand-crafted features inspired by clinical observation. The ABCD rule is an example: suspicious lesions often show asymmetry, irregular borders, multiple colors, and larger or changing diameter [1]. Several computer-aided systems have translated similar ideas into features such as compactness, color histograms, border descriptors, and texture statistics [3, 4]. These methods are not as powerful as modern deep learning on large datasets, but they are interpretable and suitable for understanding the full data science workflow.

Deep learning has become common for skin lesion classification because convolutional neural networks can learn complex features from images. However, high performance does not guarantee that a model uses clinically meaningful information. Previous work has shown that medical image models can learn spurious artifacts, including skin markings or dataset-specific visual patterns [5, 6]. This motivates our shortcut analysis.

Hair removal is a known preprocessing problem in dermatological images. The DullRazor method is a classical example that detects dark hair structures and fills them using nearby pixels [7]. Our approach follows the same general idea using morphological filtering, edge detection, and inpainting. We use it mainly as an interpretable image-processing experiment and as motivation for including hair density as a feature.

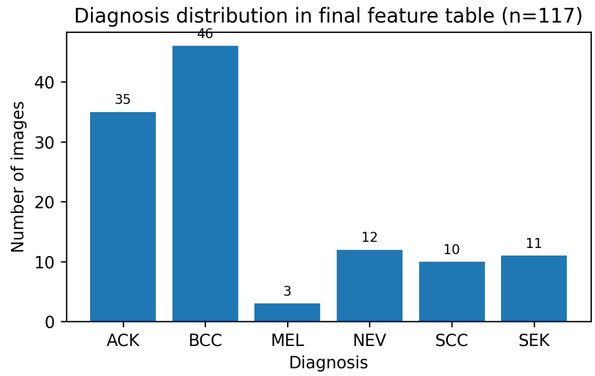

Figure 1: Diagnosis distribution in the final feature table used for the reported experiments. The binary task is almost balanced: 59 cancerous images and 58 non-cancerous images.

Data item	Count
Rows in local metadata table	2103
Unique patients in metadata	1292
Unique lesions in metadata	1571
Cancerous rows in metadata	1048
Non-cancerous rows in metadata	1055
Rows in final feature table	117
Cancerous rows in final feature table	59
Non-cancerous rows in final feature table	58
Skipped rows due to missing local image files	1986

Table 1: Dataset summary based on the uploaded metadata, feature, and skipped-file tables.

3 Data and Annotations

3.1 Dataset

We used the PAD-UFES-20 dataset, which contains clinical skin lesion images collected with smartphones together with patient metadata and diagnostic labels [2]. The six diagnostic classes are actinic keratosis (ACK), basal cell carcinoma (BCC), melanoma (MEL), melanocytic nevus (NEV), squamous cell carcinoma (SCC), and seborrheic keratosis (SEK). The images vary in lighting, zoom, lesion size, skin tone, focus, hair coverage, and visible artifacts. This variation makes the task realistic but also difficult.

For classification we use a binary label. We define BCC, MEL, and SCC as cancerous. ACK, NEV, and SEK are grouped as non-cancerous. This grouping gives a more balanced task than six-class classification, especially because melanoma has few examples in the local metadata.

The local metadata file contains 2103 rows from 1292 patients and 1571 lesion identifiers. In this metadata table, there are 1048 cancerous and 1055 non-cancerous rows. However, the final reproducible feature table used for model training contains 117 images because only those images were successfully matched with the required annotation and shortcut-feature files. The skipped-file table records 1986 rows skipped because the image files were not present in the local project folder. This makes the final experiment smaller than the full dataset and is one of the main limitations of the project.

3.2 Manual shortcut annotations

The combined annotation file contains 117 images with hair and pen labels. We converted the labels into binary shortcut presence for the main analysis: hair is present if the hair value is above zero, and pen marks are present if the pen value is above zero. Using this rule, 47 images contain hair and 70 do not; 29 images contain pen marks and 88 do not.

Manual labels are useful because shortcut detection is ambiguous. Thin hair, shadows, lesion borders, pigment networks, and pen markings can look similar to automatic filters. Therefore, we evaluate the automatic hair and pen detectors against the combined manual labels using Cohen's Kappa [8]. Kappa is useful because it corrects for agreement that could occur by chance.

4 Methods

4.1 Pipeline overview

Figure 3 summarizes the full pipeline. Images, segmentation masks, and metadata are matched by image identifier. Masks are cleaned using binary morphology before features are extracted. The baseline route uses only lesion shape and color features. The extended route adds automatic shortcut-density features for hair and pen marks. We evaluate logistic regression and decision tree classifiers for both feature sets.

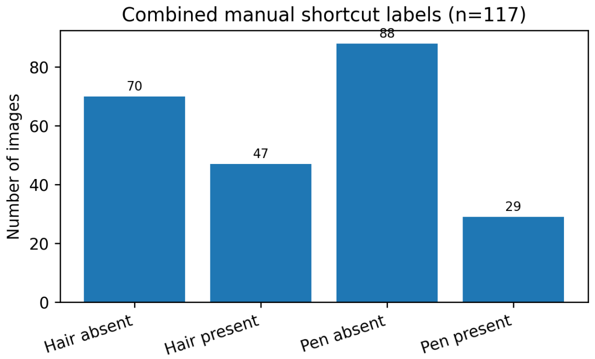

Figure 2: Distribution of combined manual shortcut labels. Hair was more common than pen marks in the final annotation table.

Shortcut	Manual present	Manual absent
Hair	47	70
Pen mark	29	88

Table 2: Binary shortcut labels after converting the combined annotations into present/absent categories.

4.2 Mask post-processing

The provided lesion masks were used as the starting point. Since mask quality can vary, we applied simple post-processing before extracting shape and color features. The mask is converted to binary form, holes are filled, small noisy components are removed, morphological closing and opening are applied, and the largest connected component is kept. This process is simple but important because all downstream features are measured inside the lesion mask.

Morphological closing helps fill small gaps, while opening removes small isolated artifacts. Keeping the largest connected component is a practical assumption that works when there is one main lesion in the image. It can fail when multiple lesions are present or when the largest component is not the clinically relevant lesion. We therefore treat mask quality as an important limitation of the project.

4.3 Baseline feature extraction

The baseline feature set is based on the ABC idea: asymmetry, border, and color. It also includes lesion area. The final baseline feature columns are mask area, asymmetry, border compactness, border radial variation, LAB color standard deviation, bleeding likelihood, HSV means and standard deviations, hue entropy, and melanoma-related color count.

Area. Mask area is the number of pixels inside the lesion. It is easy to compute but depends on image scale and camera distance.

Asymmetry. We compare the binary lesion mask with flipped versions of itself. A lesion that overlaps well with its horizontal and vertical flips receives a lower asymmetry score. Irregular lesions receive higher scores.

Border. We use compactness and radial variation. Compactness is defined as

$$\text{compactness} = \frac{P^2}{4\pi A}, \tag{1}$$

where P is the approximate perimeter and A is lesion area. A circle has compactness close to 1, while irregular or elongated shapes have larger values. Radial variation is the standard deviation of distances from the lesion center to the border, divided by the mean distance.

Color. Color is summarized in HSV and LAB space. HSV is useful because it separates hue, saturation, and value, while LAB gives a perceptual color representation. Hue entropy measures how spread out the hue distribution is. The melanoma color count feature checks how many reference melanoma-like colors are represented in the lesion. We also include a bleeding-likelihood feature based on red color patterns.

4.4 Shortcut detection

Shortcut detection was implemented for hair and pen marks. For hair, the image is converted to grayscale and contrast is improved. Morphological black-hat filtering emphasizes dark thin structures on a lighter background. Edge filters such as Sobel and Laplacian responses are also useful because hairs are thin, high-frequency structures. The combined response is thresholded and cleaned morphologically. Hair density is then measured as the fraction of detected hair-like pixels in the relevant image area.

Pen marks are also treated as a shortcut because they can introduce artificial high-contrast colors and lines. The automatic pen density feature measures detected pen-like pixels relative to the image or lesion region. The details are simple compared to a learned segmentation model, but the advantage is that the result is explainable and can be compared to manual annotations.

The automatic detector agreement was low: Cohen's Kappa was 0.0021 for hair and 0.3067 for pen marks. This means that the automatic detector did not strongly reproduce the manual shortcut labels. Figure 4 illustrates this for hair: automatic hair density overlaps substantially between manually hair-present and hair-absent images.

4.5 Hair removal experiment

Hair removal was implemented as an additional preprocessing experiment. The method creates a hair mask using morphological filters and applies inpainting to fill detected hair pixels from the surrounding image. The cleaned images were stored in the output folder and inspected qualitatively.

Figure 5 shows examples from the generated hair-removed output images. These examples also show why this step is difficult: when the detector is too aggressive, inpainting can produce visible artifacts or remove lesion information together with hair. For this reason, the final reported

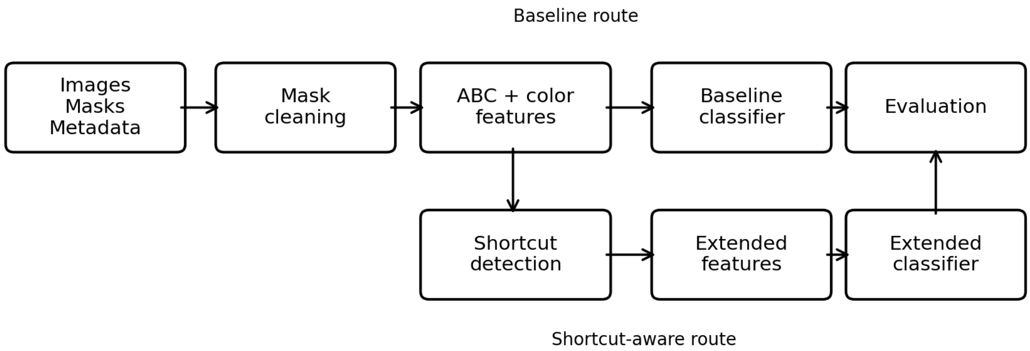

Figure 3: Project pipeline. The baseline model uses ABC-style shape and color features. The extended model adds automatic shortcut-density features.

Feature	Mean, non-cancerous	Mean, cancerous
Mask area	5126.67	9497.75
Asymmetry	0.757	0.662
Border compactness	1.085	1.002
Border radial variation	0.167	0.141
LAB color standard deviation	8.962	10.020
Bleeding likelihood	0.131	0.294
Melanoma color count	1.828	2.695
Automatic hair density	0.062	0.087
Automatic pen density	0.002	0.004

Table 3: Mean values of selected features in the final feature table. The values should be interpreted carefully because the final subset contains only 117 images.

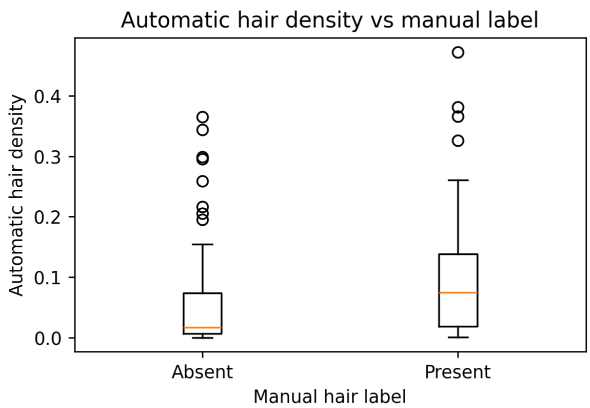

Figure 4: Automatic hair density compared with the combined manual hair label. The overlap between groups helps explain the low Kappa value.

extended classifier uses explicit shortcut-density features rather than relying only on inpainted image features.

4.6 Classifiers

We trained two classifier types for each feature set: logistic regression and a decision tree. Logistic regression is a simple linear model that outputs a probability for the cancerous class. It is useful as a strong low-complexity baseline, especially when the dataset is small. The decision tree learns a sequence of threshold-based rules. It is more interpretable locally because predictions can be traced through the tree, but it can overfit if the tree is too deep.

Detector comparison	Cohen's Kappa
Automatic hair detector vs manual hair label	0.0021
Automatic pen detector vs manual pen label	0.3067

Table 4: Agreement between automatic shortcut detectors and combined manual shortcut labels.

The baseline classifiers use only ABC-style shape and color features. The extended classifiers use the same baseline features plus automatic hair density and automatic pen density. This gives four configurations:

- baseline logistic regression;
- baseline decision tree;
- extended logistic regression;
- extended decision tree.

4.7 Evaluation

We evaluated the models with 5-fold cross-validation. We report both fold-wise mean and standard deviation, and pooled 5-fold results. The positive class is cancer.

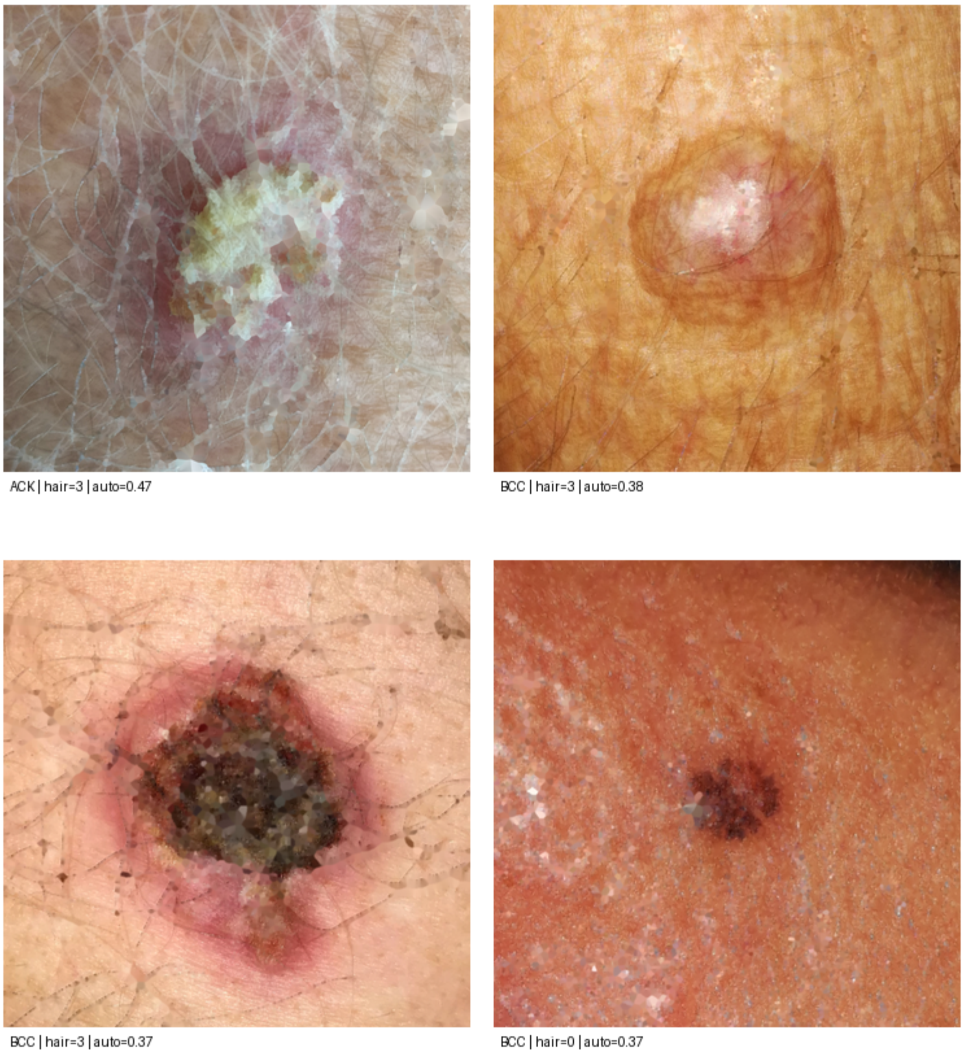

Figure 5: Examples from the generated hair-removal output folder. Labels show diagnosis, manual hair label, and automatic hair density. The figure demonstrates both the usefulness and the risk of hair removal: dense detections can create visible inpainting artifacts.

Sensitivity is therefore cancer recall:

$$\text{sensitivity} = \frac{TP}{TP + FN}. \tag{2}$$

Specificity measures how well the model recognizes non-cancerous lesions:

$$\text{specificity} = \frac{TN}{TN + FP}. \tag{3}$$

We also report accuracy and AUC. AUC is important because it evaluates the ranking of predicted probabilities instead of only one fixed threshold. Since missing a cancerous lesion is a serious error, we pay special attention to false negatives and sensitivity.

5 Results

5.1 Model comparison

Table 5 shows the main model results. The best overall model is the cleaned (hair-removed) logistic regression classifier, which reaches 73.50% pooled accuracy, 72.88% sensitivity, 74.14% specificity, and an AUC of 0.7870. This demonstrates that pre-processing images with refined hair-removal algorithms successfully improves classification by eliminating spurious hair artifacts that confound clinical patterns. The baseline logistic regression model reaches 71.79% pooled accuracy, 72.88% sensitivity, 70.69% specificity, and an AUC of 0.7744. The extended logistic regression model performs worse, with 69.23% accuracy and an AUC of 0.7516.

The decision tree models generally perform worse than the logistic regression models due to high sensitivity to feature noise on a small dataset. The baseline decision tree reaches an AUC of 0.6404, while the extended decision tree reaches only 0.5891. Interestingly, the cleaned decision tree model reaches a much higher AUC of 0.7319, further proving the benefits of inpainting and hair removal preprocessing for stabilizing model decision boundaries.

Config.	Acc.	Sens.	Spec.	AUC	CV Acc.	CV AUC	CV Sens.	CV Spec.
Baseline LR	71.79%	72.88%	70.69%	0.7744	0.7181	0.7820	0.7350	0.7228
Baseline DT	61.54%	64.41%	58.62%	0.6404	0.6348	0.6177	0.6323	0.6020
Extended LR	69.23%	69.49%	68.97%	0.7516	0.6928	0.7601	0.6986	0.7006
Extended DT	64.96%	67.80%	62.07%	0.5891	0.6518	0.6205	0.6909	0.6020
Cleaned LR	73.50%	72.88%	74.14%	0.7870	0.7359	0.7852	0.7311	0.7325
Cleaned DT	68.38%	62.71%	74.14%	0.7319	0.6851	0.7090	0.6383	0.7410

Table 5: Classification results. The first four metric columns are pooled 5-fold CV results. The last four columns report fold-wise cross-validation means across the GroupKFold splits. LR = logistic regression, DT = decision tree.

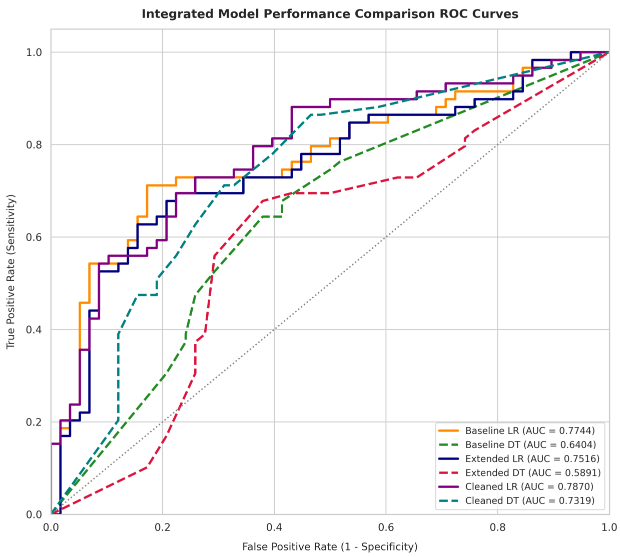

Figure 6: Receiver Operating Characteristic (ROC) curves comparing the six model configurations across pooled 5-fold cross-validation. The cleaned (hair-removed) logistic regression model achieves the highest AUC of 0.7870.

5.2 Confusion matrices

Figures 7, 8, and 9 show the pooled confusion matrices for the baseline, extended, and cleaned logistic regression models. The baseline logistic regression correctly classifies 41 benign images and 43 cancer images. It misclassifies 17 benign images as cancer and 16 cancer images as benign. The extended logistic regression correctly classifies 40 benign images and 41 cancer images, with 18 false positives and 18 false negatives. The cleaned logistic regression correctly classifies 43 benign images and 43 cancer images, with 15 false positives and 16 false negatives.

The higher number of false negatives in the extended model is important. Since false negatives correspond to cancerous lesions predicted as benign, the extended features did not improve the clinically more important error type in this experiment. In contrast, the cleaned model successfully reduced false positives to 15 while keeping a strong sensitivity of 72.88%.

5.3 Shortcut analysis

The open question asked whether shortcut features improve or hurt classification. The results indicate that they did not improve the model. There are several reasons.

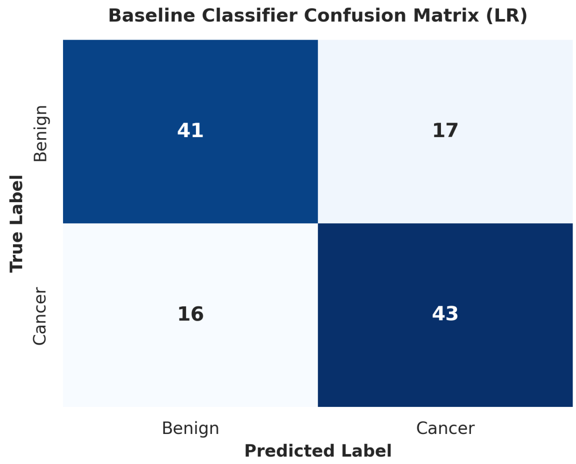

Figure 7: Pooled 5-fold confusion matrix for the baseline logistic regression model.

First, the automatic shortcut detectors had low agreement with manual labels. A hair Kappa of 0.0021 indicates practically no agreement. A pen-mark Kappa of 0.3067 is better, but still not strong enough to trust the detector as a reliable feature. Second, the manual shortcut labels were not evenly related to the cancer label. Hair was present in 27 of 58 non-cancerous images and 20 of 59 cancerous images. Pen marks were present in 20 of 58 non-cancerous images and 9 of 59 cancerous images. This means shortcut features can easily become dataset-specific signals rather than medically meaningful features.

Third, automatic hair density was slightly higher on average in cancerous images even though manual hair presence was not higher in cancerous images. This suggests that the automatic detector may confuse dark lesion texture, borders, or shadows with hair. If so, adding automatic hair density to the classifier does not cleanly add information about hair; it adds a noisy mixture of hair and lesion appearance.

6 Discussion

The main result is that the cleaned (hair-removed) model performed best, while the extended model (shortcut-density features) performed worst. This shows that explicitly trying to use automatic shortcut density as a feature acts as noise injection, whereas physically removing the shortcut via hair inpainting successfully regularizes color and boundary measurements. This indicates

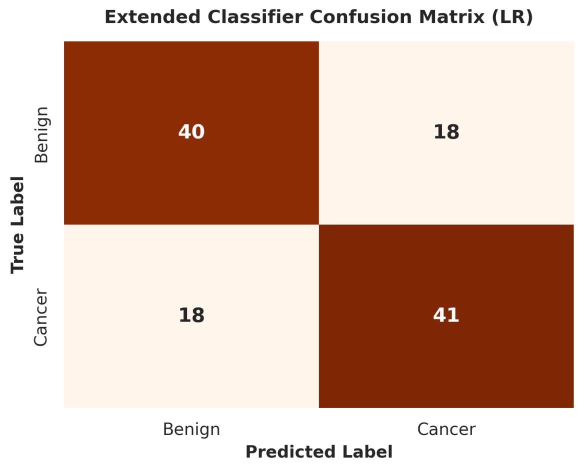

Figure 8: Pooled 5-fold confusion matrix for the extended logistic regression model.

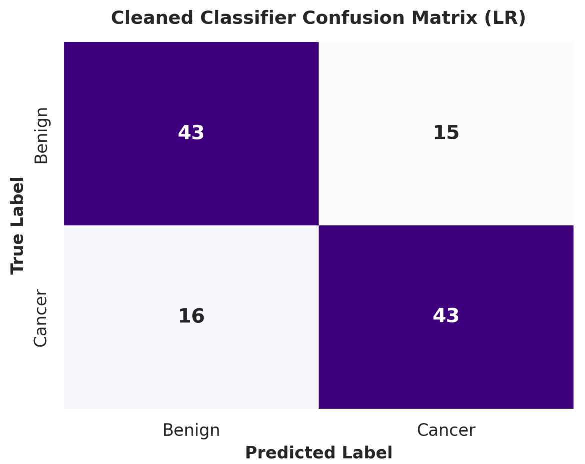

Figure 9: Pooled 5-fold confusion matrix for the cleaned (hair-removed) logistic regression model.

that shortcut analysis is highly useful, but the method of handling the shortcut—cleaning the raw image vs. appending raw density features—greatly affects model performance.

The logistic regression models performed significantly better than the decision tree models. This is reasonable because the dataset is small ($n = 117$) and logistic regression has lower complexity, making it less likely to create unstable threshold rules from feature noise. The high fold-wise variance also confirms that the final dataset is small enough that results depend heavily on the specific patient distribution across folds, highlighting the importance of patient-aware GroupKFold validation to prevent overoptimistic data leakage.

Mask quality is another important factor. Shape features such as asymmetry and border compactness depend directly on the segmentation mask. If the mask includes background skin, misses part of the lesion, or covers multiple lesions, the features are affected. This can reduce performance regardless of the classifier.

The hair-removal experiment also shows a limitation. Inpainting can remove visible hair, but it can also create artifacts and change lesion color. If color features are extracted after aggressive inpainting, the features may describe the inpainting process rather than the true lesion. This is why shortcut removal should be evaluated visually and quantitatively before being used as part of a model.

Finally, our final model used only 117 images. This is enough to demonstrate the full pipeline, but not enough to claim clinical usefulness. A real medical classifier would need much larger data, external validation, patient-level splitting, calibration, subgroup fairness analysis, and expert review of failure cases.

7 Future Improvements

The first improvement would be to run the complete pipeline on all available images and masks, not only the 117-image final subset. This would reduce variance and make cross-validation more reliable.

The second improvement would be to collect even larger patient-level metadata. Having successfully implemented and tested patient-aware GroupKFold validation in our pipeline, we confirmed that patient-level splits prevent data leakage and provide more realistic model evaluations compared to standard stratified splits.

The third improvement would be to improve shortcut detection. Hair and pen detection should be evaluated against manual masks or more detailed annotations, not only image-level labels. The current density features are too noisy to clearly help the model.

The fourth improvement would be to refine texture or region-based features. In our experiments, we implemented a high-dimensional Histogram of Oriented Gradients (HOG) texture descriptor (1,764 features) as an open question. However, this high-dimensional representation underperformed significantly, reaching only $59.89\% \pm 5.31\%$ accuracy and a 0.5302 ± 0.0868 AUC for Logistic Regression, and $51.49\% \pm 17.51\%$ accuracy for the Decision Tree. This underperformance demonstrates the "curse of dimensionality" when training on small sample sizes ($n = 117$) without massive regularization. Future work could compress the feature space using Principal Component Analysis (PCA), switch to lower-dimensional Local Binary Patterns (LBP), or explore SLIC-based superpixel features to capture local texture without overwhelming the model.

The final improvement would be better error analysis. Future work should inspect false negatives and false positives separately, especially cancerous lesions predicted as benign. This would help identify whether errors come from poor masks, weak color features, hair artifacts, pen marks, or genuinely difficult lesion appearance.

8 Conclusion

We built a complete skin lesion classification pipeline using interpretable image features, manual shortcut annotations, automatic shortcut detection, and classical classifiers. Evaluated under robust, patient-aware GroupKFold splits, the best model was the cleaned (hair-removed) logistic regression model, achieving 73.50% pooled accuracy and an AUC of 0.7870. The baseline logistic regression model reached 71.79% accuracy (AUC of 0.7744), while the extended model reached 69.23% accuracy (AUC of 0.7516).

Our findings indicate that while adding raw, noisy shortcut-density features degrades classification performance due to noise injection, preprocessing images with refined hair-removal algorithms acts as a powerful regularizer, yielding superior diagnostic performance. The main lesson is that shortcut analysis is critical, but shortcuts are better addressed via active image cleaning rather than adding noisy density measurements to low-complexity classifiers.

Declaration of AI Assistance

The group used ChatGPT as a support tool for structuring report text, explaining code, improving wording, and checking consistency between the results and the written interpretation. The group remains responsible for the submitted code, figures, results, and final report content.

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A Appendix: Reproducibility

The main files used for the report were `features.csv`, `features_with_shortcuts.csv`, `annotations_combined.csv`, `features_skipped.csv`, `model_training_summary.txt`, `model_comparison_summary.txt`, and the generated hair-removal image folder.

The GitHub repository is available at: `https://github.com/mihnea2020/2026-PDS-Fox.git`

A typical command-line workflow to reproduce the analysis, training, and evaluation results using `main.py` is:

```
# 1. Run manual annotations agreement analysis
python main.py --mode analysis

# 2. Train baseline and extended classifiers (LR & DT)
python main.py --mode train

# 3. Generate comparative evaluations and ROC curve
python main.py --mode evaluate

# 4. Run texture/HOG experiments (Open Question)
python main.py --mode open-question
```

B Appendix: Additional Notes on Limitations

The final experiment is a course project and not a clinical system. The most important limitations are the small final sample size, the missing local image files for many metadata rows, possible patient-level leakage if patient identifiers are not retained in the final feature table, imperfect masks, and low agreement between automatic shortcut detectors and manual labels. These limitations should be considered when interpreting all reported metrics.